

Question number	Answer	Additional guidance	Mark
5(a)(i)	An answer that includes the following point: • D is the correct answer • A is not the correct answer as an axon is not the space between sensory and relay neurone • B is not the correct answer as a cell body is not the space between sensory and relay neurone • C is not the correct answer as a post synaptic membrane is not the space between sensory and relay neurone		(1)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	An answer that includes three of the following points: Similarities • both transmit (electrical) impulses (1) • both may synapse with relay neurones (1) Differences • sensory neurone impulses go towards the {CNS/relay neurone} whereas motor neurone impulses go away from {CNS/relay neurone} (1) • sensory neurones transmit impulses from {receptor /sensory} cells whereas motor neurone transmit impulses to {effector/muscle} cells (1)	do not piece together – has to be in same / adjacent sentences (but not separate paragraphs) do not accept messages / signals for S1, D1 and D2 ACCEPT both release neurotransmitters (in synapse) ignore brain Note: sensory sends impulses from receptor to CNS and motor brings impulse from CNS to effector = both D1 and D2	(3)

Question number	Answer	Additional guidance	Mark
5(b)	An answer that includes three of the following points: • {loss of / no} (insulating) myelin (1) • decreases speed of nerve impulse / depolarisation no longer just occurs at nodes of Ranvier (1) • motor neurones {no longer transmit impulses / impulses are transmitted slower} (causing loss of function in the affected area) (1) • impulses from receptors do not reach CNS / impulses are not transmitted along sensory neurones to CNS (1)	ACCEPT nerve impulses take longer as has to go through all neurone not just nodes of Ranvier ACCEPT saltatory conduction does not occur ignore impulses being lost ACCEPT {fewer/slower} impulses reaching {effector / muscles} (so loss of function) (slower both mp2&3) ACCEPT fewer impulses from receptors reach CNS	(3)

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5(c)	An explanation that includes four of the following points: • a mutation changes {the base sequence (of this gene) / the amino acid sequence} (of the calcium ion channel protein) (1) • {resulting in a change in shape / prevents the opening} of the calcium ion channel (1) • this stops influx of calcium ions (across presynaptic membrane) (1) • (fewer calcium ions result in) fewer vesicles containing neurotransmitter fusing with presynaptic membrane (1) • fewer {nerve impulses/action potentials} generated at post synaptic {membrane / neurone} (1)	ACCEPT stops calcium ions passing through {membrane / ion channel} ACCEPT prevents release of neurotransmitter /less neurotransmitter will be released (into synapse) ACCEPT less depolarisation of post synaptic {membrane / neurone}	(4)